

TEST CASE PLAN

AlgoNova – AI-Powered Intelligent Learning Platform

Project	AlgoNova
Document	Software Testing – Test Case Plan
Version	1.0
Application Type	Web-based AI-powered educational platform
Technology Context	React, Node.js, Express.js, MongoDB Atlas, Google Gemini AI
Primary Modules	Authentication, Dashboard, Learning, DSA Visualizer, Coding Practice, AI Assistant, Mock Interview, System D

1. Purpose

This test case plan defines the testing approach for AlgoNova and identifies the functional, integration, usability, security, performance, and AI-related scenarios that should be validated before release. The objective is to ensure that the platform works correctly, protects user data, provides reliable learning functionality, and delivers consistent feedback across supported modules.

2. Testing Objectives

- Verify that all major features meet their specified requirements.
- Validate data flow between the React frontend, Express/Node.js backend, MongoDB Atlas, and Gemini AI services.
- Identify functional defects, integration issues, validation errors, and usability problems.
- Verify authentication, authorization, session handling, and protection of user data.
- Validate responsive behavior across desktop, tablet, and mobile screen sizes.
- Check that AI-generated responses are relevant, safe, and appropriately constrained.
- Ensure user progress, XP, badges, streaks, and assessment results are stored and displayed correctly.

3. Testing Levels and Types

Testing Type	Purpose
Unit Testing	Test individual components, functions, API handlers, and validation logic.
Integration Testing	Verify communication among frontend, backend, database, authentication, and AI services.
System Testing	Validate complete end-to-end user workflows.
UI/Usability Testing	Check navigation, layout, readability, responsiveness, and user interaction.
Security Testing	Validate authentication, authorization, input validation, session security, and API protection.
Performance Testing	Check response time, concurrent requests, API behavior, and page performance.
AI Response Testing	Evaluate relevance, correctness, consistency, hallucination risks, and inappropriate-input handling.
User Acceptance Testing	Confirm that the implemented platform satisfies intended student-learning requirements.

4. Detailed Test Cases

TC ID	Module	Test Scenario	Test Steps / Input	Expected Result	Priority
TC-001	Registration	Register with valid details	Enter valid name, email and password; submit.	Account is created and user is redirected appropriately.	High
TC-002	Registration	Reject invalid email	Enter invalid email format and submit.	Validation message is displayed; account is not created.	High
TC-003	Login	Login with valid credentials	Enter registered credentials.	User is authenticated and dashboard opens.	High
TC-004	Login	Reject incorrect password	Enter valid email with incorrect password.	Login fails with an appropriate error message.	High
TC-005	Authentication	Unauthorized protected route	Open protected page without authentication.	User is redirected to login or access is denied.	High
TC-006	Dashboard	Load user dashboard	Login and open dashboard.	Progress, XP, streak, recommendations and relevant statistics load correctly.	High
TC-007	Learning	Open a learning topic	Select a DSA topic from learning module.	Correct topic content loads without broken UI.	Medium
TC-008	Visualizer	Run algorithm visualization	Select an algorithm and start visualization.	Algorithm executes visually step-by-step with correct state transitions.	High
TC-009	Visualizer	Pause and resume visualization	Start visualization; pause; resume.	Animation pauses and continues from the correct state.	Medium
TC-010	Coding	Submit valid solution	Submit code for a supported coding problem.	Code is processed and result/status is displayed correctly.	High
TC-011	Coding	Handle invalid code	Submit syntactically incorrect code.	System reports an appropriate error without crashing.	High
TC-012	AI Assistant	Ask technical question	Enter a DSA-related question.	AI returns a relevant educational explanation.	High
TC-013	AI Assistant	Request code explanation	Provide code and request explanation.	AI explains the logic and relevant complexity appropriately.	High
TC-014	AI Review	Review submitted solution	Submit a coding solution for review.	Feedback identifies correctness, complexity, and possible improvements where applicable.	High
TC-015	Mock Interview	Start interview session	Select interview type and start.	Interview session begins and questions are presented.	High
TC-016	Mock Interview	Submit interview response	Answer an interview question and continue.	Response is recorded and next step proceeds correctly.	High
TC-017	Mock Interview	Generate interview feedback	Complete interview session.	Score/feedback and improvement areas are displayed.	High
TC-018	System Design	Create architecture	Add components and connect them in sandbox.	Components and relationships are rendered correctly.	High
TC-019	System Design	Save design	Create a design and select save.	Design is persisted and can be retrieved by the user.	Medium
TC-020	Gamification	Award XP after activity	Complete an eligible learning activity.	XP is updated exactly according to configured rules.	Medium
TC-021	Gamification	Update streak	Complete activity on an eligible consecutive day.	Learning streak is updated correctly.	Medium
TC-022	Progress	Record assessment result	Complete quiz/assessment.	Score and completion status are stored and shown correctly.	High
TC-023	Database	Persist user progress	Complete activity, refresh/re-login.	Previously stored progress remains available.	High
TC-024	Security	Reject unauthorized API request	Call protected API without valid authentication.	API denies access and does not expose protected data.	High
TC-025	Responsive UI	Test mobile layout	Open key pages at mobile viewport.	Content remains usable without major overlap or horizontal overflow.	Medium
TC-026	Error Handling	Backend/API unavailable	Simulate API failure while loading a feature.	User receives a meaningful error state and UI does not crash.	High
TC-027	AI Safety	Unsupported or malicious prompt	Send an unsafe/off-topic prompt to AI module.	System handles the prompt according to configured safety behavior and does not expose secrets.	High
TC-028	Logout	Logout from active session	Click logout and revisit protected page.	Session is terminated and protected content is inaccessible.	High

5. Test Environment

Component	Test Environment
Client	Modern Chromium/Firefox/Safari browser; desktop and mobile viewport testing
Frontend	React-based web application
Backend	Node.js + Express.js API
Database	MongoDB Atlas test database
AI Service	Google Gemini API test configuration
Network	Normal broadband/mobile network plus simulated slow/unavailable API conditions
Test Data	Dummy student accounts, sample DSA problems, sample code, interview responses and system designs

6. Entry and Exit Criteria

Entry Criteria	Exit Criteria
Requirements and major workflows are defined.	All critical and high-priority test cases have been executed.
Test environment and test data are available.	No open Critical defects remain.
Build is deployable and major dependencies are configured.	High-severity defects are fixed or formally accepted.
Required APIs and database connections are available.	Regression testing is completed for affected modules.
Test cases are reviewed before execution.	Test results and unresolved issues are documented.

7. Defect Classification

Severity	Meaning
Critical	System crash, major security failure, data loss, or core application unusable.
High	Major feature does not work or an important workflow is blocked.
Medium	Feature works with a significant issue or workaround available.
Low	Minor UI, wording, alignment, or non-critical usability issue.

8. Deliverables

- Approved test case plan
- Test case execution records
- Defect/bug reports
- Regression test results
- User acceptance test results
- Final test summary report

9. Conclusion

The proposed test case plan provides structured coverage for AlgoNova's major functional and non-functional requirements. Testing should be performed incrementally throughout development rather than only at the end of the project. Particular attention should be given to authentication, persistence of learner progress, AI integrations, coding workflows, system design interactions, and error handling because these areas involve multiple components and external dependencies.